

Protocoll: Experiment 15: Inclined Plane

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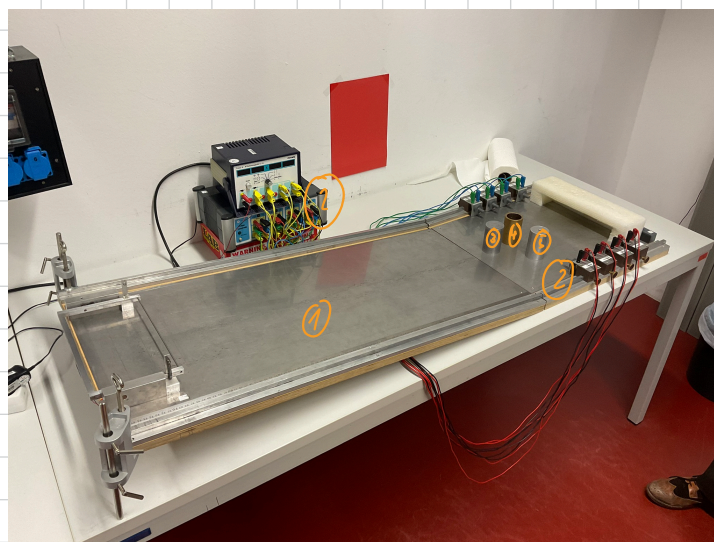


Fig 3.1 Setup

Setup :

- Height-adjustable runway (1)
- solid cylinder: alu , $\rho = 2,70 \frac{\text{g}}{\text{cm}^3}$ (3)
- hollow cylinder: brass , $\rho = 8,44 \frac{\text{g}}{\text{cm}^3}$ (4)
- composite cylinder: sheath of aluminum, core of brass (5)
- Calliper
- Scale
- Spirit level

A1: Measurement

Tab 3.1: Measurement of rolling bodies

cylinder	d _{inner} [mm]	d _{outer} [mm]	mass [g]
solid	0	$5,00 \pm 0,05$	$443,0 \pm 1,0$
hollow	$4,65 \pm 0,05$	$5,00 \pm 0,05$	$443,0 \pm 1,0$
(composite)	$0,76 \pm 0,05$	$5,00 \pm 0,05$	$443 \pm 1,0$

length of track [cm]: $100,0 \pm 0,5 \text{ cm}$

With the error estimation based on the error of the ruler and the uncertainty of the measurement method (the plane the ruler is set on is not perfectly flat and there is a tiny gap between the measured length and the actual plane (wood))

$$\Delta \sigma = \frac{\sigma}{\sqrt{n}} \quad \Delta C = 0,05 \text{ cm}$$

We adjust for the error of the calliper not being upright by doing multiple measurements.

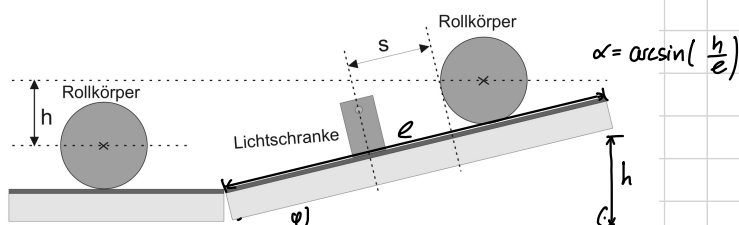
$$\text{Total error: } \Delta = \sqrt{\Delta \sigma^2 + \Delta C^2}$$

Height of track [cm]:

- 1: 11,10
- 2: 12,80
- 3: 11,50
- 4: 12,80
- 5: 12,70

with $\Delta \sigma = 0,37$

$$\bar{h} = (12,2 \pm 0,4) \text{ cm}$$



The time error for all time measurements is $\Delta t = \pm 0,0015$

A2: movement type? consistent acceleration

$$L_i = L_{\text{end}} - L_{\text{start}}, \Delta r_{\text{ruler}} = 0,05 \text{ cm}$$

Tbl 3.2 Full cylinder

barrier [cm] Δr_{tot}	time [s]
$l_1 = 25,00 \pm 0,07$	$t_1 = 0,921$
$l_2 = 36,00 \pm 0,07$	$t_2 = 1,084$
$l_3 = 49,00 \pm 0,07$	$t_3 = 1,272$
$l_4 = 64,00 \pm 0,07$	$t_4 = 1,451$

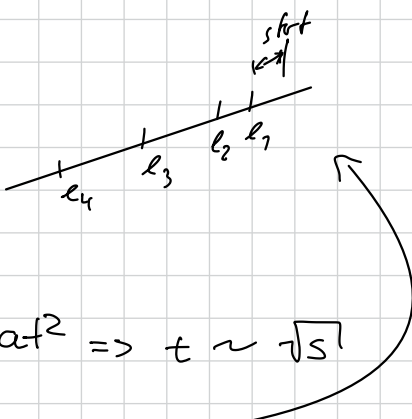
The uncertainty of the light source not being perfectly centered is neglected in the face of the much bigger uncertainty of the ruler. Nevertheless it is worth for a mention.

We aligned both sides (sender and receiver) with the ruler so no alignment error aside from Δr is expected.

We add the two separate Δr :

$$\Delta r_{\text{tot}} = \sqrt{\Delta r_1^2 + \Delta r_2^2} = \sqrt{2} \cdot 0,05 =$$

Δr_1 : sender side, Δr_2 : receiver side

$$s = \frac{1}{2}at^2 \Rightarrow t \sim \sqrt{s}$$


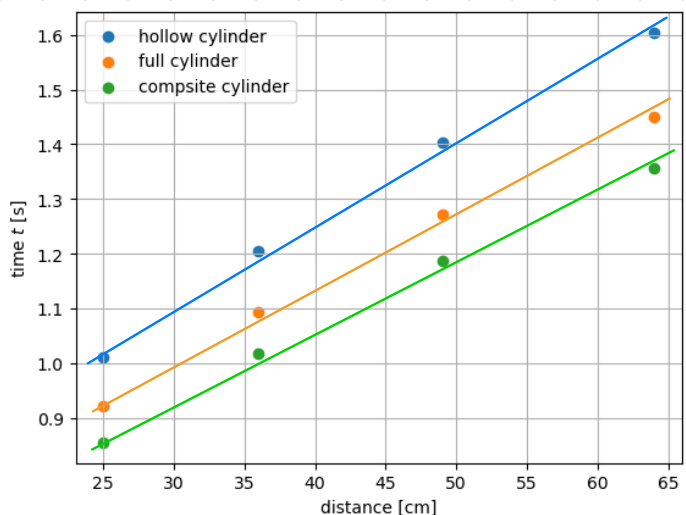
Hollow cylinder

barrier [cm] Δr_{tot}	time [s]
$l_1 = 25,00 \pm 0,07$	$t_1 = 1,011$
$l_2 = 36,00 \pm 0,07$	$t_2 = 1,204$
$l_3 = 49,00 \pm 0,07$	$t_3 = 1,403$
$l_4 = 64,00 \pm 0,07$	$t_4 = 1,604$

Composed cylinder

barrier [cm] Δr_{tot}	time [s]
$l_1 = 25,00 \pm 0,07$	$t_1 = 0,855$
$l_2 = 36,00 \pm 0,07$	$t_2 = 1,018$
$l_3 = 49,00 \pm 0,07$	$t_3 = 1,186$
$l_4 = 64,00 \pm 0,07$	$t_4 = 1,356$

Plots:



Tabl: 3.3 Measurement of acceleration

A3 [cm] dist $\Delta t_{off} = 0,07$ time hollow $\Delta t =$ [ms] t_{full} [ms]

$l_1 = 25$	1,019	1,027	1,014	1,022	1,009	0,922	0,914	0,800	0,801	0,913
$l_2 = 36$	1,212	1,223	1,209	1,215	1,201	1,096	1,089	1,075	1,074	1,087
$l_3 = 49$	1,410	1,426	1,410	1,415	1,400	1,278	1,268	1,255	1,254	1,266
$l_4 = 64$	1,610	1,629	1,613	1,614	1,600	1,458	1,446	1,435	1,435	1,446

Table 3.4
Mean + standard error:

	Hollow [s]	Full [s]
l_1 :	$1,018 \pm 0,004$	$0,906 \pm 0,006$
l_2 :	$1,212 \pm 0,004$	$1,080 \pm 0,005$
l_3 :	$1,412 \pm 0,004$	$1,260 \pm 0,006$
l_4 :	$1,613 \pm 0,005$	$1,440 \pm 0,006$

A4:

```

: # messung, t1, t2

# hollow
m1h = np.array([1.922, 2.192])
m2h = np.array([1.951, 2.222])
m3h = np.array([1.936, 2.206])
m4h = np.array([1.922, 2.192])
m5h = np.array([1.938, 2.207])

# full
m1f = np.array([1.724, 1.967])
m2f = np.array([1.736, 1.980])
m3f = np.array([1.718, 1.961])
m4f = np.array([1.737, 1.981])
m5f = np.array([1.749, 1.992])

```

TbL 3.5

Measurement of translational velocity Hollow:

[cm]	Distance	Δt_{tot}	[s] ΔT_1	[s] ΔT_2	[s] ΔT_3	[s] ΔT_4	[s] ΔT_5
$\Delta L =$	$25,00 \pm 0,07$		0,270	0,271	0,270	0,270	0,269

$$\overline{T}_H = (0,2700 \pm 0,0010)\text{s} \text{ with } \Delta \overline{T}_H = \sqrt{\left(\frac{\sigma_{\overline{T}_H}}{\sqrt{5}}\right)^2 + \Delta t^2} \Rightarrow \overline{v}_H = \frac{\Delta L}{\overline{T}_H}$$

TbL 3.6

Measurement of translational velocity Full:

[cm]	Distance	Δt_{tot}	[s] ΔT_1	[s] ΔT_2	[s] ΔT_3	[s] ΔT_4	[s] ΔT_5
$\Delta L =$	$25,00 \pm 0,07$		0,243	0,244	0,243	0,244	0,243

$$\overline{T}_F = (0,2430 \pm 0,0010)\text{s} \text{ with } \Delta \overline{T}_F = \sqrt{\left(\frac{\sigma_{\overline{T}_F}}{\sqrt{5}}\right)^2 + \Delta t^2} \Rightarrow \overline{v}_F = \frac{\Delta L}{\overline{T}_F}$$

Error propagation!



img 3.2 Distance

Wg
(b/a/26)



$$h = \sin(\alpha) \cdot (l - 11\text{mm})$$

V35

September 16, 2026

height of track

```
[5]: import numpy as np
import matplotlib.pyplot as plt
```

```
[2]: h = np.array([11.1, 12.8, 11.5, 12.8, 12.7])
err_h = 0.05

print(np.mean(h))
error_mean = np.std(h, ddof=1)/np.sqrt(len(h))
print(error_mean)
print(np.sqrt(error_mean**2 + err_h**2))
```

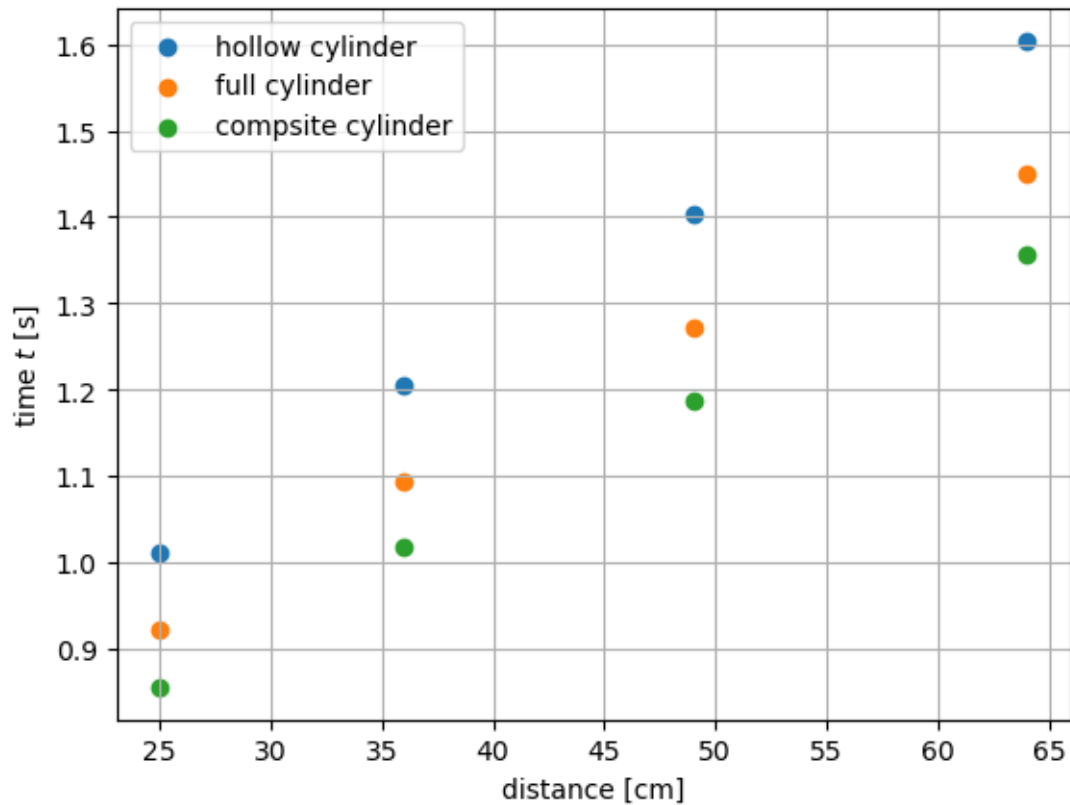
```
12.180000000000001
0.3652396473549936
0.3686461718233353
```

```
[ ]: t_hollow = np.array([1.011, 1.204, 1.403, 1.604])
t_full = np.array([0.921, 1.094, 1.272, 1.451])
t_compose = np.array([0.855, 1.018, 1.186, 1.356])

x = np.array([25, 36, 49, 64])

plt.scatter(x, t_hollow, label="hollow cylinder")
plt.scatter(x, t_full, label="full cylinder")
plt.scatter(x, t_compose, label="composite cylinder")
plt.xlabel("distance [cm]")
plt.ylabel("time $t$ [s]")
plt.grid(True)
plt.legend()
```

```
[ ]: <matplotlib.legend.Legend at 0x11bb9cb90>
```



Quantitative A3:

```
[13]: x = np.array([25, 36, 49, 64])

t_hollow = np.array([
    [1.019, 1.212, 1.410, 1.610],
    [1.027, 1.223, 1.426, 1.629],
    [1.014, 1.209, 1.410, 1.613],
    [1.022, 1.215, 1.415, 1.614],
    [1.009, 1.201, 1.400, 1.600]
])
t_full = np.array([
    [0.922, 1.096, 1.278, 1.459],
    [0.893, 1.068, 1.245, 1.425],
    [0.900, 1.075, 1.255, 1.435],
    [0.901, 1.074, 1.254, 1.435],
    [0.913, 1.087, 1.266, 1.446]
])

mean_hollow = np.mean(t_hollow, axis=0)
```

```

std_hollow = np.std(t_hollow, axis=0, ddof=1)/np.sqrt(t_hollow.shape[0]) # 
    ↳Standard error of the mean
mean_full = np.mean(t_full, axis=0)
std_full = np.std(t_full, axis=0, ddof=1)/np.sqrt(t_full.shape[0]) # Standard 
    ↳error of the mean

print("Hollow cylinder:")
for distance, mean, std in zip(x, mean_hollow, std_hollow):
    print(f"{distance} cm: mean = {mean:.4f} s, std = {std:.4f} s")

print("\nFull cylinder:")
for distance, mean, std in zip(x, mean_full, std_full):
    print(f"{distance} cm: mean = {mean:.4f} s, std = {std:.4f} s")

```

Hollow cylinder:

```

25 cm: mean = 1.0182 s, std = 0.0031 s
36 cm: mean = 1.2120 s, std = 0.0036 s
49 cm: mean = 1.4122 s, std = 0.0042 s
64 cm: mean = 1.6132 s, std = 0.0047 s

```

Full cylinder:

```

25 cm: mean = 0.9058 s, std = 0.0052 s
36 cm: mean = 1.0800 s, std = 0.0050 s
49 cm: mean = 1.2596 s, std = 0.0057 s
64 cm: mean = 1.4400 s, std = 0.0058 s

```

a4

```

[20]: # messung, t1, t2

# hollow
m1h = np.array([1.922, 2.192])
m2h = np.array([1.951, 2.222])
m3h = np.array([1.936, 2.206])
m4h = np.array([1.922, 2.192])
m5h = np.array([1.938, 2.207])

# full
m1f = np.array([1.724, 1.967])
m2f = np.array([1.736, 1.980])
m3f = np.array([1.718, 1.961])
m4f = np.array([1.737, 1.981])
m5f = np.array([1.749, 1.992])

m_diff_h = np.array([])
for i in [m1h, m2h, m3h, m4h, m5h]:
    m_diff_h = np.append(m_diff_h, i[1] - i[0])

```

```

m_diff_f = np.array([])
for i in [m1f, m2f, m3f, m4f, m5f]:
    m_diff_f = np.append(m_diff_f, i[1] - i[0])

err_t = 0.001
print("Hollow cylinder time differences:")
print(m_diff_h)
print("Mean time difference for hollow cylinder:", np.mean(m_diff_h))
print("Standard error for hollow cylinder:", np.std(m_diff_h, ddof=1)/
      ↪len(m_diff_h))
print('Full error for hollow cylinder:', np.sqrt((np.std(m_diff_h, ddof=1)/
      ↪len(m_diff_h))**2 + err_t**2))

print("\nFull cylinder time differences:")

print(m_diff_f)
print("Mean time difference for full cylinder:", np.mean(m_diff_f))
print("Standard error for full cylinder:", np.std(m_diff_f, ddof=1)/
      ↪len(m_diff_f))
print('Full error for full cylinder:', np.sqrt((np.std(m_diff_f, ddof=1)/
      ↪len(m_diff_f))**2 + err_t**2))

```

```

Hollow cylinder time differences:
[0.27  0.271 0.27  0.27  0.269]
Mean time difference for hollow cylinder: 0.27000000000000001
Standard error for hollow cylinder: 0.00014142135623730964
Full error for hollow cylinder: 0.0010099504938362077

Full cylinder time differences:
[0.243 0.244 0.243 0.244 0.243]
Mean time difference for full cylinder: 0.2434
Standard error for full cylinder: 0.00010954451150102927
Full error for full cylinder: 0.001005982107196743

```